

2. FREQUENCY & PITCH

HOW DO WE DISCUSS A MUSICAL TONE? (EXCLUDING ATTACK, DECAY, RELEASE)
(HUM EXAMPLES, DIFFERENT PITCH & TIMBRE)

Q MATCHING GAME:

MUSICIANS

PHYSICISTS

PITCH

FREQUENCY

INTENSITY (ENERGY PER SECOND PER AREA = $\frac{W}{m^2} = \frac{J/s}{m^2}$)

LOUDNESS

AMPLITUDE (WAVE PEAK DISPLACEMENT, VELOCITY, PRESSURE, ...)

POWER (ENERGY PER SECOND = $W = INTENSITY \times AREA$)

TIMBRE

OVERTONE STRUCTURE (OTHER, HIGHER FREQUENCIES)

HARMONICS (INTEGER MULTIPLES OF THE FUNDAMENTAL (LOWEST) FREQUENCY)

"WATT" = J/s



- SECONDARY CONNECTIONS:

- PERCEIVED LOUDNESS IS FREQUENCY-DEPENDENT (CAN'T HEAR 40kHz, e.g.)
- HARMONICS ADD POWER & CAN INFLUENCE PERCEIVED LOUDNESS
- FREQUENCY & AMPLITUDE CAN CHANGE THE PERCEIVED TIMBRE

- ALSO DYNAMICS AFFECT THESE PERCEPTIONS

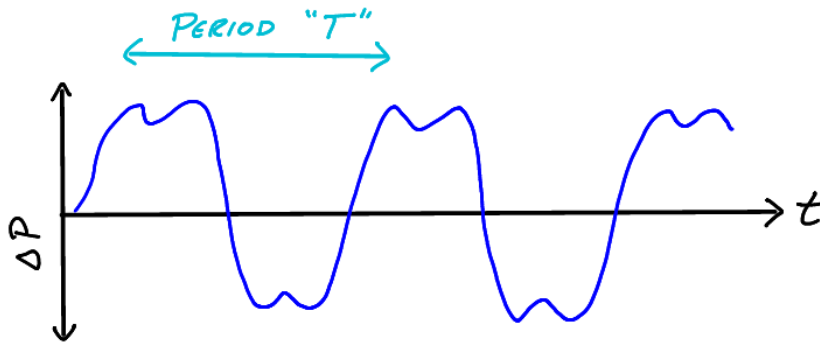
DEMO

RECORD SOME WAVEFORMS & LOOK FOR PERIODIC STRUCTURE (I USE AUDACITY FOR THIS):

- CLAP (DISCUSS AXES)
- GENERATED 220 Hz
- STUDENT TRY TO MIMIC, CHECK SHAPE
- MIMIC THEN INCREASE PITCH TO SEE FREQUENCY INCREASE
- NORMAL VOICE

- PERIODIC WAVES HAVE A DEFINITE "PITCH" OR "FUNDAMENTAL" (LOWEST) FREQUENCY.
- MATHEMATICALLY AND AS WE PERCEIVE THEM

FREQUENCY + PERIOD: QUANTITIES WE'VE USED ALREADY BUT HAVEN'T CAREFULLY CONSIDERED



$\Delta P = \text{CHANGE IN PRESSURE RELATIVE TO THE MEAN.}$
 $= P(t) - \bar{P}$

- THE "PERIOD" (T) OF AN OSCILLATION IS THE TIME IT TAKES TO REPEAT (UNITS = $s/\text{cycle} = s$)
- THE (FUNDAMENTAL) "FREQUENCY" (f) IS THE INVERSE PERIOD (UNITS OF $\text{cycles}/s = 1/s = \text{Hz}$)

$$f = \frac{1}{T} \quad \text{FREQUENCY + PERIOD}$$

- EXAMPLE: "MODERATO" TEMPO OF 100 BEATS PER MINUTE:

$$T = 60 \text{ seconds} / 100 = 0.6 \text{ seconds}$$

$$\Rightarrow f = 1/T = 1.67 \text{ Hz}$$

- EXAMPLE: 440 Hz TUNING FORK ("A" ON A PIANO)

$$\Rightarrow T = 2.27 \text{ ms} = 0.00227 \text{ s}$$

PITCH PERCEPTION

- NATURAL HEARING RANGE: 20 Hz - 20 kHz (GIVE OR TAKE)

- IF A PULSE OF SOUND REPEATS WITH FREQUENCY BELOW $\sim 20 \text{ Hz}$, YOUR BRAIN INTERPRETS THEM AS INDIVIDUAL PULSES. ABOVE $\sim 20 \text{ Hz}$, IT BEGINS TO HEAR PITCH

END OF
MON 9-9

PRODUCT OF EVOLUTION?

- ANIMALS OUR SIZE CANNOT MOVE FASTER THAN $\sim 20 \text{ Hz}$
- SIMILAR TO OUR VISUAL PERCEPTION (COINCIDENCE?)
 - TRADITIONAL MOVIE IMAGES = 24 Hz ← CAN SEE ARTIFACTS. WE LIKE THESE ARTIFACTS, IT SEEMS
 - LOWER FRAME RATES LOOK "CHOPPY"
- GOAL IN VIDEO GAMES $\sim 120 \text{ Hz}$
(MORE ABOUT FAST PANNING + STROBE AT THIS RATE)

LIONS DON'T ATTACK AT 20 Hz! PEOPLE DON'T PUNCH THIS FAST!
(ALSO, 20 Hz - 20 kHz MATCHES OUR SPEECH... OR VIS VERSA?)

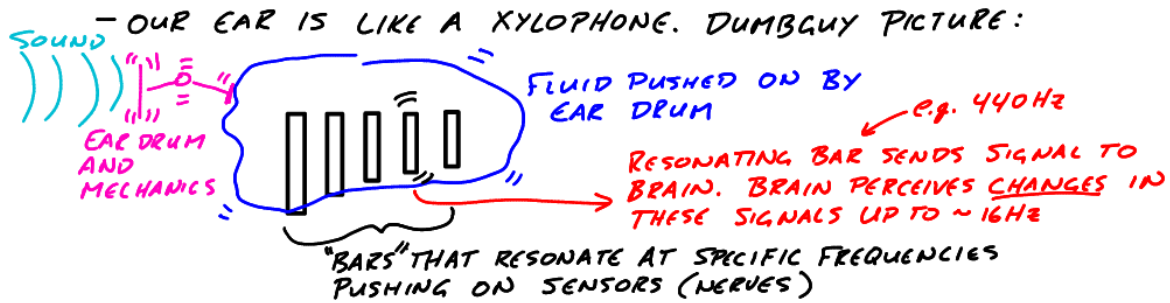
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CLICKS AT DIFFERENT RATES

- INSPECT WITH SOUND EDITOR

- AT WHAT FREQUENCY DO YOU HEAR A TONE?



10pt

Q

AT WHAT FREQUENCY DOES THE BIGGEST BAR RESONATE?

- LOW-FREQUENCY RANGE (≤ 50 Hz)

- EAR SENSITIVITY IS DECREASING

- MORE OF A THING YOU FEEL (IN CHEST + BODY, TOO!)

- NOTE MOST LOW-FREQUENCY TONES/INSTRUMENTS HAVE ADDITIONAL OVERTONES MIXED IN, MAKING THEM EASIER TO RECOGNIZE, E.G., THE CLICKS DEMO.

DEMO

PURE TONE, 32 Hz (SOUND EDITOR, SPEAKER)

- NOTE THE SPEAKER LIKELY GOES "NONLINEAR" GENERATING 64 Hz, 128 Hz, ... (WEAKLY), MAKING THIS LESS PURE / EASIER TO HEAR

- NOTE THE SPEAKER MAY NOT RESPOND MUCH AT 32 Hz, TOO (IT CAN'T GENERATE AS MUCH POWER)

"FREE" FUN: PLAY 7 Hz PURE TONE THROUGH A NICE WOOFER. PEOPLE WILL FEEL A SENSE OF DREAD. USED IN HORROR MOVIES AND GAMES (MY QUEST 2?)

- HIGH FREQUENCY RANGE: USUALLY UP TO ~ 17 kHz.

DEMO GENERATE TONES IN SOUND EDITOR

11 kHz

12 kHz

⋮

VOTE WHO CAN HEAR WHAT.

(NOTE THE SPEAKERS MAY NOT BE GOOD ENOUGH.)

^{10pt}

Q ANYONE ELSE HEAR THE NOISE FLOOR (HISS) IN THEIR EARS?

PITCH

IN MUSIC, WE DEVELOP A SENSE OF HOW MUCH "HIGHER" OR "LOWER" ONE NOTE IS THAN ANOTHER. BUT WHAT IS A NATURAL SCALE FOR THIS (IS THERE ONE)?

^{Opt}

Q WHICH SET OF INTERVALS SOUND/FEEL "EVENLY SPACED"?

- MOST WILL AGREE THAT, OVER THE FULL RANGE, "EQUAL RATIO" SOUNDS THE MOST "EVENLY" SPACED.

POSSIBLE CAUSES:

FOR SURE **DEMO** MAJOR SCALE SOUNDS "EVEN" WHEN IT'S NOT!

- WE ARE ALL BIASED BY THE MUSIC WE HEAR

- OUR EAR XYLOPHONE HAS THIS KIND OF SPACING

- THERE IS SOMETHING MATHEMATICALLY BEAUTIFUL ABOUT THIS, ALSO.

→ ALL THREE, I BELIEVE, BUT THE EAR XYLOPHONE DOES HAVE ROUGHLY "EQUAL RATIO" BAR FREQUENCIES.

⇒ HUMANS BASICALLY MADE XYLOPHONES IN THE IMAGE OF THEIR EAR SENSORS WITHOUT KNOWING WHY? ☺

→ THERE IS MATHEMATICAL ELEGANCE IN THE SCALES WE CHOSE FOR "MUSIC"... MORE LATER (HOPEFULLY!)

- THIS "EQUAL RATIO" BUSINESS IS ALL WE MEAN WHEN WE SAY SOMETHING IS "LOGARITHMICALLY SPACED"

WHY? LOOK AT THE n^{th} FREQUENCY:

$$f_n = f_0 B^n$$

LOGARITHMIC SPACING

↑ INDEX INTEGER ↑ STARTING FREQUENCY SOME CONSTANT, e.g. 1.3

$$\Rightarrow f_0 = f_0$$

$$f_1 = f_0 \times B$$

$$f_2 = f_1 \times B = f_0 \times B \times B$$

$$f_3 = f_2 \times B = f_0 \times B \times B \times B$$

⋮

$$\Rightarrow \log f_0 = \log f_0$$

$$\Rightarrow \log f_1 = \log f_0 + \log B$$

$$\Rightarrow \log f_2 = \log f_0 + 2 \log B$$

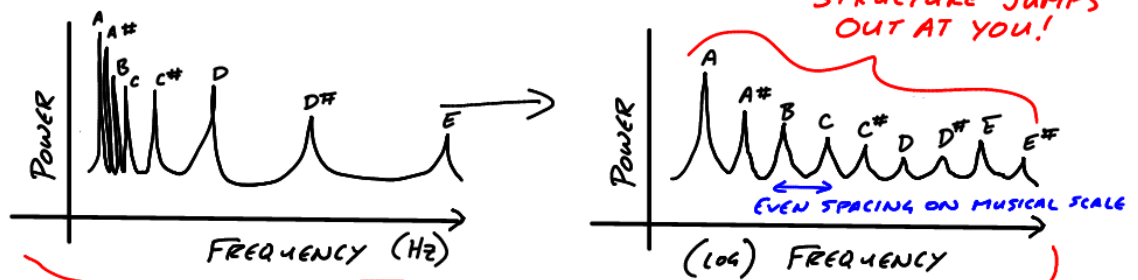
$$\Rightarrow \log f_3 = \log f_0 + 3 \log B$$

⋮

WE WILL DISCUSS THIS NEXT!

EVEN SPACING ON LOG SCALE

- THIS IS WHY AUDIO SPECTRA ARE SO OFTEN PLOTTED ON A "LOG-SPACED" FREQUENCY SCALE.



SAME SPECTRA PLOTTED DIFFERENTLY

10pt

Q WHAT IS "B" FOR OUR 12-NOTES-PER OCTAVE SCALE?
THE CONSTANT, NOT THE NOTE!

SUMMARY

$$f = \frac{1}{T}$$

FREQUENCY =
REPETITIONS PER SECOND

FREQUENCY + PERIOD

PERIOD =
TIME BETWEEN REPETITIONS

- WE HEAR FROM $\sim 20\text{Hz}$ - 20kHz
ON A LOGARITHMIC FREQUENCY SCALE

- WHAT SOUNDS "EVENLY SPACED"

$$f_n = f_0 \beta^n$$

SOME CONSTANT, E.G. 1.3 OR $\sqrt[12]{2}$

LOGARITHMIC SPACING

STARTING FREQUENCY
INDEX INTEGER